王裕宸 2024201594

设 $|U| = u$, $|B| = b$

令映射到桶 $c \in B$ 的元素个数为 $n_c = |\{x \in U : h(x) = c\}|$

$\forall (x, y) \in U \times U$ 且 $x \neq y$ ，定义 $I_{x,y}(h) = \begin{cases} 1 & h(x) = h(y) \\ 0 & h(x) \neq h(y) \end{cases}$

$\therefore N_{碰撞} = \sum_{x \neq y} I_{x,y}(h) = \sum_{c \in B} n_c(n_c - 1)$

$\therefore E[\sum_c n_c(n_c-1)] = \sum_{x \neq y} P[h(x) = h(y)] \leq u(u-1)\varepsilon$ ①

$\because$ 查阅资料有 $\sum_c n_c^2 \geq \frac{u^2}{b}$

$\therefore \sum_c n_c(n_c-1) \geq \frac{u^2}{b} - u = u(\frac{u}{b} - 1)$

$\Rightarrow E[\sum_c n_c(n_c-1)] \geq u(\frac{u}{b} - 1)$ ②

由 ① ② 可得 $u(\frac{u}{b} - 1) \leq u(u-1)\varepsilon$

$$\Rightarrow \varepsilon \geq \frac{u-b}{b(u-1)} \geq \frac{u-b}{bu} = \frac{1}{b} - \frac{1}{u}$$

$\therefore \varepsilon \geq \frac{1}{|B|} - \frac{1}{|U|}$ 得证